

Silence-the-Mimic: Accelerating Imperceptible Perturbation Generation Against Voice Cloning

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Abstract—Deep neural network-based Voice Conversion (VC) and Text-to-Speech (TTS) models have rapidly advanced, enabling realistic voice cloning with minimal input data. Such capabilities raise serious concerns over unauthorized cloning of speaker identities and the associated privacy and security risks. Current imperceptible adversarial protection methods rely on quality control losses that are highly sensitive to hyperparameter tuning and computationally expensive due to lengthy optimization. To address these limitations, we propose a fast protection method that generates perceptually constrained perturbations in the frequency domain under a psychoacoustic masking-based constraint. Our approach strictly enforces perceptibility bounds during adversarial training, eliminating the need for iterative quality balancing and significantly reducing computational cost. Experiments on multiple state-of-the-art VC and TTS models show that STM achieves competitive or superior protection performance with substantially better perceptual quality and up to 45.3× speedup over existing white-box baselines. These results demonstrate the effectiveness of frequency-domain perturbations with perceptual constraints as a practical paradigm for protecting against voice cloning.

Index Terms—adversarial defense, audio security, voice cloning, voice conversion, text-to-speech.

I. INTRODUCTION

Recent Voice Conversion (VC) and Text-to-Speech (TTS) models require only a few seconds of a target speaker’s audio to generate high-fidelity synthetic speech while allowing arbitrary linguistic content [1], [2]. Despite their beneficial applications, highly realistic synthesized speech, commonly referred to as deepfake audio, poses serious threats to individuals’ privacy, property, and reputation.

In response to these threats, existing research has explored methods to defend against unauthorized deepfake audio [3], [4]. These approaches can be broadly categorized into two groups: *detection* and *protection*. Deepfake audio detection focuses on distinguishing machine-generated speech from human speech. This can be achieved by examining artifacts present in synthesized audio (passive detection) [5] or by extracting watermarks embedded in the audio beforehand (active detection) [6]. However, detection operates only after an attack has occurred. In contrast, protection methods add imperceptible perturbations before publication to disrupt speaker-identity extraction and prevent unauthorized cloning [7].

This paper proposes a new methodology that is significantly faster than existing imperceptible white-box adversarial-perturbation-based protection methods. Existing adversarial protection methods typically combine the attack objective with

auxiliary waveform-, spectral-, or perception-based quality losses. However, their performance is sensitive to the relative quality-loss weight, and hundreds or thousands of iterations are often required to balance protection and audio quality [8], [9], limiting practical deployment.

Our proposed method addresses these limitations by introducing imperceptible perturbations in the frequency domain. Specifically, we first convert the speech signal into a spectrogram in the frequency domain using the Short-Time Fourier Transform (STFT), under the Constant Overlap Add (COLA) condition to ensure stable inverse STFT (iSTFT) reconstruction. A psychoacoustic model is then applied to the spectrogram to estimate the masking threshold of each time-frequency bin in every STFT frame. This threshold is combined with a predefined tolerance to establish a strictly enforced human perceptibility constraint, ensuring that the injected perturbations remain within the allowable range. In this manner, gradient updates are directed solely toward the perturbations, which accelerates adversarial training while preserving strong protection performance.

We conduct extensive experiments on multiple state-of-the-art VC and TTS models and compare our approach with leading white-box defense baselines. The results demonstrate that the proposed method achieves comparable or superior protection performance while substantially accelerating perturbation generation across all evaluated settings.

II. RELATED WORK

Voice Cloning Models. Voice cloning refers to generating speech with arbitrary linguistic content that mimics the vocal characteristics of a specific individual. Recent approaches mainly fall into two paradigms: VC [10]–[16] and TTS [2], [17]–[22]. VC models take utterances from two speakers: one provides linguistic content (what to say), and the other provides speaker identity (how to say it). The model then combines these inputs to synthesize speech that carries the content of the former while preserving the vocal traits of the latter. TTS models, on the other hand, generate acoustic features from input text conditioned on a short reference audio that provides speaker information. These features are then passed to a vocoder to synthesize speech that preserves the vocal characteristics of the reference speaker.

Detection-based Protection. To mitigate the risks posed by unauthorized voice cloning, researchers have developed proactive algorithms that embed stealth identifiers into audio

files to enable post-cloning tracking, a technique commonly referred to as audio watermarking [6], [23]–[25]. Audio watermarking methods typically involve a generator–detector framework: the generator embeds imperceptible patterns into speech, which tend to remain intact even after processing or cloning by VC or TTS models, while the detector recovers or verifies these patterns to trace the audio’s origin and assess its authenticity.

Another approach to detecting deepfake audio is to analyze intrinsic artifacts in model synthesized speech [5]. These passive methods identify acoustic, spectral, or temporal inconsistencies introduced during synthesis and use statistical models or deep neural networks to distinguish synthetic speech from genuine recordings.

Adversarial Perturbation Protection. Another promising defense strategy is to embed imperceptible or minimally perceptible noise into speech audio [8], [9], [26]–[29]. Such noise is specifically designed to disrupt the cloning process, particularly the speaker encoder, thereby preventing voice cloning models from synthesizing accurate outputs. Adversarial-perturbation-based defenses can be categorized along two binary dimensions: how the perturbations are generated and where they are applied. With respect to generation, some methods produce perturbations through adversarial training, while others employ a pretrained perturbation generator that does not require gradient information during inference. With respect to application, perturbations are either injected directly into the audio waveform or introduced by biasing latent representations within a vocoder.

III. METHODOLOGY

In this section, we present the proposed protection algorithm, Silence-the-Mimic (STM). Unlike prior approaches that rely on perceptual loss functions to regulate the audibility of perturbations, STM employs a psychoacoustic-based hard-clamping strategy to enforce strict control over perceptual distortion in the frequency domain. This design enables direct control over the perceptual quality of protected speech while simultaneously accelerating the perturbation process by eliminating the need for additional audio refinement steps.

We begin this section by formulating the protection problem, followed by a detailed description of our methodology.

A. Problem Setup

Our problem scenario involves three entities: (i) a victim who publishes publicly available content containing their speech samples, denoted by the set $\mathbb{X}_u$, (ii) an attacker who seeks to exploit these samples to generate deepfake audio, and (iii) a legitimate verifier authorized to authenticate the victim.

The attacker is modeled as a third-party adversary without the capability to directly capture the victim’s speech. Instead, the attacker collects publicly available speech samples of the victim. To generate spoofed speech, the attacker employs a speech synthesis model $G : \mathbb{R}^{d_{enc}} \times \mathcal{C} \rightarrow \mathbb{R}^*$, which synthesizes an utterance $G(enc_{\text{spk}}(\mathbf{x}), c)$ containing linguistic content c while mimicking the vocal characteristics of the

target utterance $\mathbf{x} \in \mathbb{X}_u$. Here, $\mathbf{x} \in \mathbb{R}^*$ denotes a variable-length sequence representing the audio waveform, where the symbol $*$ indicates that the sequence length may vary. $c \in \mathcal{C}$ denotes arbitrary content selected by an attacker to facilitate misuse, and $enc_{\text{spk}} : \mathbb{R}^* \rightarrow \mathbb{R}^{d_{enc}}$ denotes the speaker encoder.

The verifier relies on a speaker verification (SV) model $SV : \mathbb{R}^* \times \mathbb{R}^* \rightarrow [0, 1]$ to compute the similarity between an enrollment utterance from the victim and a trial utterance. The verifier accepts the trial utterance as authentic if the similarity score exceeds a predefined threshold, thereby serving as the ultimate gatekeeper in controlling access to victim-authorized interactions.

To safeguard the victim’s voice identity from unauthorized replication, a protection algorithm perturbs each utterance $\mathbf{x} \in \mathbb{X}_u$ before publication, so that any speech cloned from the protected utterance $\mathbf{x} + \Delta\mathbf{x}$ is rejected by the SV model when compared with the victim’s enrollment utterance $\mathbf{x}_{\text{enr}}$. At the same time, the perturbation is constrained to remain perceptually unobtrusive, ensuring that the protected utterance remains intelligible and usable. This process can be formulated as:

$$\begin{aligned} \min_{\Delta\mathbf{x}} \quad & SV(\mathbf{x}_{\text{enr}}, G(enc_{\text{spk}}(\mathbf{x} + \Delta\mathbf{x}), c)) \\ \text{s.t.} \quad & H(\mathbf{x} + \Delta\mathbf{x}, \mathbf{x}) \leq \mathbf{H}_{\text{limit}}, \end{aligned} \quad (1)$$

where $H : \mathbb{R}^* \times \mathbb{R}^* \rightarrow \mathbb{R}^d$ denotes a human perceptual function that quantifies the dissimilarity between two inputs, and the output dimensionality d may correspond to a scalar or a high-dimensional tensor depending on the granularity of perceptual constraints (e.g., time–frequency bins). The parameter $\mathbf{H}_{\text{limit}}$ specifies the maximum allowable distortion and enables fine-grained control across different feature channels. In practice, directly optimizing Equation 1 is often infeasible. Instead, we modify the speaker identity perceived by the synthesizer to achieve a similar outcome, as discussed in a later section.

B. Method Formulation

Our proposed method, STM, begins by converting the victim’s speech audio into a frequency-domain representation, i.e., a spectrogram. A psychoacoustic model is then applied to constrain the maximum allowable distortion of the protected audio. Finally, the optimization problem in Equation 1 is reformulated into an empirical form suitable for adversarial training, resulting in the generation of the protected output. The overall STM pipeline is shown in Figure 1.

Frequency-domain Perturbation. We perform adversarial protection by introducing perturbations in the frequency domain. The raw speech waveform is converted into a spectrogram using STFT. In particular, we employ a Hann window with a 50% overlap between consecutive frames. This configuration satisfies the COLA condition, ensuring that the sum of overlapping windows remains constant over time. The COLA property guarantees that the signal can be reliably reconstructed through iSTFT, which is critical for our framework, as perturbed spectrograms must be transformed back

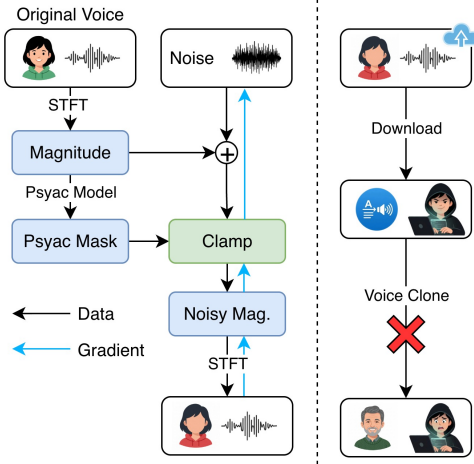


Fig. 1. Pipeline of STM.

into time-domain waveforms without introducing additional artifacts.

We denote the STFT and iSTFT operations by $STFT : \mathbb{R}^* \rightarrow \mathbb{R}^{* \times *}$ and $iSTFT : \mathbb{R}^{* \times *} \rightarrow \mathbb{R}^*$, respectively, and write the spectrogram of $x \in \mathbb{X}_u$ as $\mathbf{Y}_{\text{spec}} = STFT(x)$. Here for notational simplicity, we retain only the magnitude component and omit the phase, which remains unchanged throughout the algorithm. We denote the corresponding frequency-domain perturbation as Δ_{spec} , which shares the same shape as $\mathbf{Y}_{\text{spec}}$.

Approximate Perceptibility Constraint. To directly control the audibility of added noise, it is essential to quantitatively approximate the distortion perceived by humans between the raw speech audio and its protected version. In STM, we employ a psychoacoustic model, denoted as $Psyac(\mathbf{Y}_{\text{spec}}) : \mathbb{R}^{* \times *} \rightarrow \mathbb{R}^{* \times *}$, to estimate the masking threshold. This threshold, combined with predefined tolerance margins, provides an approximate representation of the human perceptual constraint in Equation 1.

Specifically, let $\mathbf{Y}_{\text{mask}} = Psyac(\mathbf{Y}_{\text{spec}})$ denote the masking threshold of the spectrogram $\mathbf{Y}_{\text{spec}}$. We then identify audible and inaudible time-frequency bins in the original utterance as follows:

$$\mathbf{1}_{\text{audible}} = \mathbf{1}_{\geq \mathbf{Y}_{\text{mask}}}(\mathbf{Y}_{\text{spec}}), \quad \mathbf{1}_{\text{inaudible}} = \mathbf{1}_{< \mathbf{Y}_{\text{mask}}}(\mathbf{Y}_{\text{spec}}),$$

where $\mathbf{1}_{\text{condition}}(\cdot)$ is an element-wise indicator function. We then impose the approximate perceptibility constraint on the protected audio as:

$$\begin{aligned} \mathbf{Y}_{\text{spec}}^{\text{adv}} &\leq \mathbf{Y}_{\text{spec}} + \mathbf{1}_{\text{audible}} \odot h_{\text{audible}}^{\text{head}} + \mathbf{1}_{\text{inaudible}} \odot h_{\text{inaudible}}^{\text{head}} := \mathbf{H}^{\text{head}}, \\ \mathbf{Y}_{\text{spec}}^{\text{adv}} &\geq \mathbf{Y}_{\text{spec}} + \mathbf{1}_{\text{audible}} \odot h_{\text{audible}}^{\text{floor}} + \mathbf{1}_{\text{inaudible}} \odot h_{\text{inaudible}}^{\text{floor}} := \mathbf{H}^{\text{floor}}, \end{aligned} \quad (2)$$

where $\mathbf{Y}_{\text{spec}}^{\text{adv}} = \mathbf{Y}_{\text{spec}} + \Delta_{\text{spec}}$ denotes the spectrogram of the protected utterance, $\odot$ represents element-wise multiplication, and $h_{\{\text{audible}, \text{inaudible}\}}^{\{\text{head}, \text{floor}\}}$ correspond to the headroom and floorroom (lower-bound) tolerances for audible and inaudible time-frequency bins, respectively. These tolerance values are tunable parameters of STM that explicitly control the trade-off between audio quality and protection strength. For instance, by setting

$$h_{\text{audible}}^{\text{floor}} = 0, \quad h_{\text{audible}}^{\text{head}} = 0, \quad h_{\text{inaudible}}^{\text{floor}} = -\infty, \quad h_{\text{inaudible}}^{\text{head}} = \mathbf{Y}_{\text{mask}} - \mathbf{Y}_{\text{spec}},$$

STM enforces a fully masked perturbation under the adopted psychoacoustic model, where audible bins remain unchanged and inaudible bins are constrained strictly below the masking threshold.

Alternatively, by allowing controlled distortion through:

$$h_{\text{audible}}^{\text{floor}} = -c, \quad h_{\text{audible}}^{\text{head}} = c, \quad h_{\text{inaudible}}^{\text{floor}} = -\infty, \quad h_{\text{inaudible}}^{\text{head}} = \mathbf{Y}_{\text{mask}} - \mathbf{Y}_{\text{spec}} + c,$$

STM relaxes the constraint on audible bins by a margin $c > 0$, thereby increasing the protection strength at the cost of a small, controlled reduction in perceptual quality.

Hard Clamping. An effective approach used in this study to enforce the perceptibility constraint in Equation 2 is to hard clamp excessive values of $\mathbf{Y}_{\text{spec}}^{\text{adv}}$ (and thus of Δ_{spec}) back within the headroom and floorroom tolerances at the end of each training iteration:

$$\text{Clamp}(\mathbf{Y}_{\text{spec}}^{\text{adv}}) = \min(\max(\mathbf{Y}_{\text{spec}}^{\text{adv}}, \mathbf{H}^{\text{floor}}), \mathbf{H}^{\text{head}}),$$

where min and max are applied element-wise, returning the smaller or larger value of their two operands, respectively.

Targeted Embedding Shift. The ultimate objective of our problem is to reduce the confidence score produced by the SV model. However, directly optimizing the objective function in Equation 1 is inefficient and, in the case of autoregressive voice cloning models, often infeasible. Instead, we optimize the perturbation Δ_{spec} to alter the output of the speaker encoder, thereby lowering the SV confidence score.

In prior work on white-box adversarial protection methods, two primary objectives have been employed to deviate the output of the speaker encoder: *threshold-based* and *target-based*. The threshold-based approach trains the perturbation to push the speaker encoder's output a sufficient distance away from the original speaker embedding under a chosen distance metric, with the protection rate controlled by a predefined threshold. In contrast, the target-based approach forces the perturbed embedding to resemble that of another speaker selected from an embedding bank. In STM, we adopt the target-based approach, as guiding perturbations toward a valid subspace of real speech embeddings consistently yields higher audio quality in the protected outputs. Additional analysis of this effect is presented in Section IV-F.

Multi-Step Total Variation Loss. To further enhance the temporal smoothness of perturbations and thereby improve audio quality, we incorporate a *multi-step* one-dimensional total variation (TV) loss into the optimization objective. Rather than penalizing only differences between adjacent time frames, this formulation considers multiple step sizes to capture variations across different temporal scales. Formally, the multi-step TV loss is defined as:

$$\text{Loss}_{\text{MTV}} = \sum_{k \in \mathcal{K}} \frac{1}{T-k} \sum_{t=1}^{T-k} |\Delta_{\text{spec}}[:, t+k] - \Delta_{\text{spec}}[:, t]|,$$

where $\mathcal{K}$ is the set of step sizes, and T denotes the length of Δ_{spec} along the time axis. By enforcing smoothness across both short- and long-range time intervals, this regularization reduces rapid oscillations and stabilizes the perturbation structure.

The complete STM protection procedure is summarized in Algorithm 1.

Algorithm 1: Silence-the-Mimic (STM)

Require: speaker encoder enc_{spk} ; STFT $STFT$; inverse STFT $iSTFT$; psychoacoustic model $Psyac$; clamp function $Clamp$; params $h_{\{\text{head}, \text{floor}\}}^{\{\text{audible}, \text{inaudible}\}}$; loss $Loss$

Input: victim utterance $\mathbf{x}$

Output: protected utterance $\mathbf{x}_{\text{adv}}$

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1:  $\mathbf{Y}_{\text{spec}} \leftarrow STFT(\mathbf{x})$ ,  $\mathbf{Y}_{\text{mask}} \leftarrow Psyac(\mathbf{Y}_{\text{spec}})$ 
2:  $emb_{\text{spk}}^{\text{orig}} \leftarrow enc_{\text{spk}}(\mathbf{x})$ 
3: Choose  $emb_{\text{spk}}^{\text{trg}}$  from bank using  $emb_{\text{spk}}^{\text{orig}}$ 
4: Compute  $\mathbf{H}^{\text{head}}$ ,  $\mathbf{H}^{\text{floor}}$  via Equation 2
5: Initialize  $\Delta_{\text{spec}}$ ; initialize optimizer  $\mathcal{O}$ 
6: for  $t \leftarrow 1$  to  $T_{\text{iter}}$  do
7:    $\mathbf{Y}_{\text{spec}}^{\text{adv}} \leftarrow \mathbf{Y}_{\text{spec}} + \Delta_{\text{spec}}$ 
8:    $\mathbf{x}_{\text{adv}} \leftarrow iSTFT(\mathbf{Y}_{\text{spec}}^{\text{adv}})$ 
9:    $emb \leftarrow enc_{\text{spk}}(\mathbf{x}_{\text{adv}})$ ,
       $loss \leftarrow Loss(emb, emb_{\text{spk}}^{\text{trg}})$ ,
       $\nabla_{\Delta_{\text{spec}}} \leftarrow \partial loss / \partial \Delta_{\text{spec}}$ 
10:   $\Delta_{\text{spec}} \leftarrow \mathcal{O}\text{-update}(\Delta_{\text{spec}}, \nabla_{\Delta_{\text{spec}}})$ 
11:   $\Delta_{\text{spec}} \leftarrow$ 
       $Clamp(\Delta_{\text{spec}}; \mathbf{Y}_{\text{spec}}, \mathbf{Y}_{\text{mask}}, \mathbf{H}^{\text{head}}, \mathbf{H}^{\text{floor}})$ 
12:  $\mathbf{x}_{\text{adv}} \leftarrow iSTFT(\mathbf{Y}_{\text{spec}} + \Delta_{\text{spec}})$ 
13: return  $\mathbf{x}_{\text{adv}}$ 

```

IV. EXPERIMENTS

A. Dataset

We evaluate our proposed protection method and baseline methods on the CSTR VCTK Corpus [30]. For preprocessing, we first discard utterances without transcripts, and then randomly select 100 speakers, with 5 utterances randomly chosen from each, resulting in a total of 500 utterances. From the remaining utterances, we select the longest one from each speaker to construct the embedding bank for target-based adversarial training. All audio files are downsampled to 16 kHz prior to training or inference with any model.

B. Target Models and Baselines

To best resemble a powerful attacker trying to clone the voice identity of the victim, we select four popular open-source DNN-based voice synthesis models with state-of-the-art zero-shot or few-shot voice cloning capabilities. The selected models are: GPT-SoVITS (GSV [22]), FreeVC [14], QuickVC [15] and TriAAN-VC [16].

We evaluate the defense performance of our proposed method on the models described above, comparing it with two baselines: AntiFake [8] and VoiceGuard [9]. Both methods are widely recognized and continue to serve as baselines in recent studies [7], [31], underscoring their status as current leading approaches. They operate under a white-box setting and employ adversarial training to generate perturbations in the time domain, combined with loss-based audio quality control.

The key distinction lies in their strategies for quality control: AntiFake integrates the loss terms directly into the perturbation process, whereas VoiceGuard performs audio refinement as a separate stage following adversarial training, with the optimal weight determined via binary search.

C. Evaluation Metrics

We evaluate the protection effectiveness of STM and the baseline methods using three widely adopted SV models: TitaNet [32], ECAPA-TDNN [33], and Pyannote.audio [34], [35]. For TitaNet and ECAPA-TDNN, we employ the implementations provided in NVIDIA’s NeMo library, while for Pyannote.audio, we use the implementation from its official repository.

Following prior work, we evaluate the audio quality of protected utterances using the Mean Opinion Score (MOS) and the Noisiness metric. NISQA-predicted MOS provides an overall measure of perceived speech quality, while the Noisiness metric quantifies degradation due to additive noise, capturing the extent to which an audio sample is perceived as “not noisy” [36], [37]. In addition, we incorporate the Perceptual Evaluation of Speech Quality (PESQ) [38] to assess perceptual speech quality, and the Short-Time Objective Intelligibility (STOI) [39] to evaluate speech intelligibility. Together, these metrics provide a comprehensive evaluation by capturing complementary aspects of perceptual quality, ranging from naturalness to intelligibility and noise robustness. MOS and Noisiness are obtained using NISQA [37], an open-source DNN-based model for multidimensional speech quality prediction. For PESQ and STOI, we adopt the `pesq`¹ and `pystoi`² libraries, respectively.

D. Training Details

For our proposed method, we configure the STFT with an FFT size of 512 and a hop size of 256 to satisfy the COLA condition. The resulting spectrograms are subsequently converted to a logarithmic scale. The psychoacoustic model is adopted from [40]. We extract the speaker encoder components from GSV, FreeVC, QuickVC, and TriAAN-VC, and employ them as enc_{spk} in Algorithm 1. The target speaker embedding, $emb_{\text{spk}}^{\text{trg}}$, is selected as the embedding corresponding to the fifth-largest distance from the original speaker embedding, $emb_{\text{spk}}^{\text{orig}}$, in the entire speaker bank. Perceptibility control parameters are set as follows:

$$h_{\text{audible}}^{\text{floor}} = -0.15, \quad h_{\text{audible}}^{\text{head}} = 0.15, \quad h_{\text{inaudible}}^{\text{floor}} = -\infty, \quad h_{\text{inaudible}}^{\text{head}} = \mathbf{Y}_{\text{mask}} - \mathbf{Y}_{\text{spec}}.$$

We use the Adam optimizer with a learning rate of 1 and parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$. T_{iter} is set to 80. For the loss function, we use a combination of L1 loss and cosine similarity loss as shown in Equation 3:

$$Loss(emb, emb_{\text{spk}}^{\text{trg}}) = \left\| emb - emb_{\text{spk}}^{\text{trg}} \right\|_1 + \left(1 - \text{CosineSimilarity}(emb, emb_{\text{spk}}^{\text{trg}}) \right). \quad (3)$$

¹<https://github.com/ludlows/PESQ>

²<https://github.com/mpariente/pystoi>

TV loss is incorporated into Equation 3 for defense against FreeVC and QuickVC.

For the baselines, we evaluate AntiFake under three configurations: 80, 500 and 1000 iterations. The 1000-iteration configuration corresponds to its official setting, while the 80-iteration configuration matches STM’s optimization budget for a fair comparison. We also manually adjust the weight ratio between the speaker loss and the quality control loss to balance audio fidelity and defense robustness. Additional evaluations of AntiFake with different quality control loss weights are provided in Section IV-F. For VoiceGuard, we adopt the configuration from the original paper, consisting of 3000 adversarial training steps followed by 1500 refinement steps.

For the three SV models, the thresholds of normalized cosine similarity are set to 0.7318, 0.6466, and 0.7507 for ECAPA-TDNN, Pyannote, and TitaNet, respectively. These thresholds are determined by their Equal Error Rates (EER) on the VCTK dataset.

All methods are evaluated on a single NVIDIA A100 80GB PCIe GPU. Training time is averaged over the entire sampled dataset, accounting only for perturbation iterations and excluding data I/O.

E. Main Results

Table I reports the comparative SV rejection rates and audio quality metrics. TitaNet, E-TDNN, and Pyannote denote the rejection rates obtained from the TitaNet, ECAPA-TDNN, and Pyannote.audio SV systems, respectively. F-VC, Q-VC, and T-VC denote the target models FreeVC, QuickVC, and TriAAN-VC, respectively. STM denotes our proposed method, AF-80, AF-500 and AF-1000 correspond to AntiFake with 80, 500 and 1000 iterations, and VG represents VoiceGuard. Confidence intervals are reported at the 95% level. The symbol $\uparrow$ indicates that higher values are preferable. The best results are shown in **bold**, and the second best results are underlined.

Across all VC and TTS models, STM consistently demonstrates competitive protection performance. It achieves the highest or second-highest rejection rates on nearly all SV systems, frequently surpassing VoiceGuard and AntiFake by large margins. With protection performance comparable to or exceeding that of the baselines, STM preserves superior perceptual quality in the protected audio. It achieves competitive MOS and PESQ scores, while its STOI and Noisiness results are consistently higher than those of the baselines, indicating more effective perceptual masking of perturbations.

As shown in Table II, STM consistently achieves the lowest protection runtime across all target models. At the matched 80-iteration budget, STM is still $1.59\times$ faster than AntiFake on average (2.19 vs. 3.48 s/utterance), while substantially larger speedups are observed against AntiFake under its official 1000-iteration setting, as well as against VoiceGuard. This demonstrates that STM’s runtime advantage persists even when controlling for the optimization budget.

TABLE I
COMPARISON OF SV REJECTION RATES AND AUDIO QUALITY METRICS BETWEEN THE PROPOSED METHOD AND THE BASELINE METHODS.

(a) SV rejection rates and MOS.					
Model	Method	TitaNet $\uparrow$	E-TDNN $\uparrow$	Pyannote $\uparrow$	MOS $\uparrow$
GSV	STM	96.00%	95.40%	68.80%	3.26 $\pm$ 0.06
	AF-80	93.20%	91.20%	80.00%	2.80 $\pm$ 0.06
	AF-500	83.20%	86.40%	66.80%	2.98 $\pm$ 0.04
	AF-1000	80.00%	72.60%	48.40%	3.25 $\pm$ 0.06
	VG	<u>94.00%</u>	<u>95.20%</u>	81.40%	2.82 $\pm$ 0.05
F-VC	STM	98.40%	98.80%	94.40%	3.80 $\pm$ 0.07
	AF-80	89.20%	88.00%	<u>78.60%</u>	2.88 $\pm$ 0.06
	AF-500	76.80%	77.60%	70.40%	3.22 $\pm$ 0.05
	AF-1000	76.20%	77.20%	69.40%	3.26 $\pm$ 0.11
	VG	<u>83.20%</u>	<u>84.40%</u>	76.00%	2.99 $\pm$ 0.04
Q-VC	STM	99.60%	99.60%	98.80%	3.64 $\pm$ 0.08
	AF-80	91.60%	92.20%	87.20%	2.91 $\pm$ 0.08
	AF-500	57.20%	56.80%	51.60%	3.22 $\pm$ 0.04
	AF-1000	51.00%	51.20%	41.20%	3.55 $\pm$ 0.04
	VG	<u>97.80%</u>	<u>97.80%</u>	<u>95.40%</u>	2.90 $\pm$ 0.04
T-VC	STM	67.80%	65.60%	<u>43.40%</u>	2.86 $\pm$ 0.06
	AF-80	74.40%	71.80%	39.40%	2.86 $\pm$ 0.09
	AF-500	67.40%	<u>68.00%</u>	46.00%	2.86 $\pm$ 0.04
	AF-1000	67.40%	67.20%	46.00%	2.98 $\pm$ 0.03
	VG	<u>69.20%</u>	67.60%	42.80%	<u>2.89 $\pm$ 0.04</u>
(b) Audio quality metrics.					
Model	Method	Noisiness $\uparrow$	PESQ $\uparrow$	STOI $\uparrow$	
GSV	STM	3.65 $\pm$ 0.04	2.54 $\pm$ 0.03	0.91 $\pm$ 0.00	
	AF-80	1.73 $\pm$ 0.02	1.81 $\pm$ 0.05	0.81 $\pm$ 0.02	
	AF-500	1.89 $\pm$ 0.02	1.86 $\pm$ 0.03	0.79 $\pm$ 0.01	
	AF-1000	2.03 $\pm$ 0.04	2.00 $\pm$ 0.04	<u>0.82 $\pm$ 0.01</u>	
	VG	1.78 $\pm$ 0.02	1.72 $\pm$ 0.02	0.78 $\pm$ 0.01	
F-VC	STM	3.82 $\pm$ 0.05	2.86 $\pm$ 0.03	0.94 $\pm$ 0.00	
	AF-80	1.91 $\pm$ 0.02	1.67 $\pm$ 0.05	0.78 $\pm$ 0.03	
	AF-500	2.12 $\pm$ 0.03	1.95 $\pm$ 0.03	<u>0.81 $\pm$ 0.01</u>	
	AF-1000	2.30 $\pm$ 0.09	2.04 $\pm$ 0.06	0.80 $\pm$ 0.03	
	VG	1.93 $\pm$ 0.02	1.79 $\pm$ 0.02	0.78 $\pm$ 0.01	
Q-VC	STM	3.67 $\pm$ 0.06	2.59 $\pm$ 0.03	0.95 $\pm$ 0.00	
	AF-80	1.76 $\pm$ 0.01	1.63 $\pm$ 0.05	0.80 $\pm$ 0.02	
	AF-500	2.15 $\pm$ 0.01	2.20 $\pm$ 0.03	0.83 $\pm$ 0.01	
	AF-1000	<u>2.45 $\pm$ 0.02</u>	<u>2.58 $\pm$ 0.03</u>	<u>0.86 $\pm$ 0.01</u>	
	VG	1.74 $\pm$ 0.01	1.73 $\pm$ 0.02	0.77 $\pm$ 0.01	
T-VC	STM	3.54 $\pm$ 0.03	2.36 $\pm$ 0.03	0.89 $\pm$ 0.01	
	AF-80	1.76 $\pm$ 0.01	1.80 $\pm$ 0.05	0.81 $\pm$ 0.03	
	AF-500	1.81 $\pm$ 0.01	1.87 $\pm$ 0.03	0.79 $\pm$ 0.01	
	AF-1000	1.91 $\pm$ 0.01	<u>2.03 $\pm$ 0.03</u>	0.80 $\pm$ 0.01	
	VG	<u>1.92 $\pm$ 0.01</u>	1.97 $\pm$ 0.03	<u>0.83 $\pm$ 0.01</u>	

F. Further Experiments

Robustness Against Purification Attacks. We evaluate whether common audio purification operations can remove STM perturbations before voice conversion. Following AntiFake [8], we apply four representative preprocessing operations to protected reference utterances before FreeVC inference: 8-bit quantization-dequantization, 8 kHz down/up-sampling, frequency filtering, and Griffin–Lim mel-spectrogram inversion. These operations cover both mild

TABLE II
PROTECTION RUNTIME ACROSS TARGET MODELS AND METHODS.

Model	Runtime ↓ (s/utterance)				
	STM	AF-80	AF-500	AF-1000	VG
GSV	1.02	<u>2.60</u>	18.35	38.34	30.87
F-VC	2.01	<u>3.09</u>	20.10	43.38	56.80
Q-VC	1.76	<u>3.12</u>	21.02	42.55	79.72
T-VC	3.96	<u>5.09</u>	27.85	57.53	156.27

TABLE III
ROBUSTNESS AGAINST PURIFICATION ATTACKS ON FREEVC.

Metric	None	Quant.	Resamp.	Freq. Filtering	Mel Inversion
Rejection Rate ↑	98.80%	99.60%	96.20%	91.60%	91.20%

waveform-level transformations and stronger spectral-domain reconstruction attacks. The converted speech is then evaluated using the same ECAPA-TDNN verifier. For frequency filtering, we use shelf filters to attenuate frequencies below $0.1C$ and above $1.5C$ by 30 dB, where C is the utterance-level median spectral centroid.

As shown in Table III, STM remains robust to all purification operations, maintaining rejection rates above 90% even under the stronger frequency-filtering and Griffin–Lim reconstruction attacks. These results indicate that standard purification can attenuate STM perturbations but does not reliably remove the protection in this setting.

Black-box Transferability. We evaluate the transferability of our method following the black-box protocol provided in AntiFake’s official implementation. Specifically, we optimize perturbations against three surrogate models, SV2TTS (RTVC) [41], AdaptiveVC [11], and YourTTS [42], using the same evaluation set as in our main experiments, and test their effectiveness on the unseen closed-source commercial system ElevenLabs. STM achieves an average SV rejection rate of 71.50%, a NISQA-predicted MOS of 3.37, and a PESQ of 2.47, compared with 81.80%, 2.30, and 1.22 for AntiFake, respectively. Although AntiFake achieves a higher rejection rate, STM retains substantial black-box protection while providing considerably better audio quality.

Effect of Quality Control Loss Weight in AntiFake. In Table I, AntiFake is reported under the manually selected quality control loss weights used for the main comparison. To better characterize the sensitivity of AntiFake to this hyperparameter, we further evaluate AntiFake-500 across a broader range of loss weight ratios, with the corresponding results presented in Table IV. Column *QC Weight* specifies the applied adjustment to the quality control loss. The notation $\times c$ indicates that the loss weight is scaled by the factor c , while a value of 1 denotes direct substitution of the target model’s speaker encoder in AntiFake. For FreeVC, a QC weight of $\times 0.5$ is equivalent to drop-in and is therefore not reported.

As shown in the table, for GSV, FreeVC, and QuickVC, when AntiFake’s protection rate approaches that of STM, its audio quality degrades substantially compared to STM. Moreover, as the weight of the quality control loss increases, the protection rate decreases even further relative to STM, un-

TABLE IV
EFFECT OF VARYING THE QUALITY CONTROL LOSS WEIGHT IN ANTIFAKE.

Model	QC weight	TitaNet ↑	E-TDNN ↑	Pyannote ↑	MOS ↑	Noisiness ↑	PESQ ↑	STOI ↑
GSV	$\times 2$	68.60%	73.40%	47.80%	2.95 ± 0.04	1.99 ± 0.02	2.07 ± 0.03	0.82 ± 0.01
	$\times 0.5$	96.40%	96.60%	89.60%	2.85 ± 0.04	1.69 ± 0.02	1.59 ± 0.02	0.75 ± 0.01
	1 (drop-in)	100.00%	100.00%	99.60%	2.02 ± 0.03	1.46 ± 0.01	1.09 ± 0.00	0.60 ± 0.01
F-VC	$\times 5$	2.76%	3.31%	5.52%	3.80 ± 0.05	2.86 ± 0.06	2.99 ± 0.05	0.89 ± 0.01
	$\times 2$	54.60%	55.20%	47.00%	3.32 ± 0.04	2.41 ± 0.02	2.30 ± 0.04	0.84 ± 0.01
	$\times 0.5$	—	—	—	—	—	—	—
	1 (drop-in)	95.80%	95.00%	90.60%	1.57 ± 0.03	1.37 ± 0.01	1.22 ± 0.01	0.69 ± 0.01
Q-VC	$\times 2$	53.00%	54.00%	46.40%	3.22 ± 0.04	2.17 ± 0.01	2.23 ± 0.03	0.83 ± 0.01
	$\times 0.5$	63.00%	62.40%	55.20%	3.26 ± 0.04	2.10 ± 0.01	2.15 ± 0.03	0.83 ± 0.01
	1 (drop-in)	97.00%	97.20%	93.20%	2.02 ± 0.03	1.46 ± 0.01	1.09 ± 0.00	0.60 ± 0.01
	$\times 2$	99.60%	99.80%	98.00%	1.49 ± 0.02	1.42 ± 0.01	1.13 ± 0.01	0.66 ± 0.01
T-VC	$\times 0.5$	90.20%	90.60%	76.60%	2.60 ± 0.04	1.72 ± 0.01	1.70 ± 0.02	0.76 ± 0.01
	1 (drop-in)	100.00%	99.80%	100.00%	1.12 ± 0.01	1.47 ± 0.01	1.07 ± 0.00	0.58 ± 0.01

TABLE V
EVALUATION OF STM WITHOUT DIRECTED EMBEDDING SHIFT.

Model	TitaNet ↑	E-TDNN ↑	Pyannote ↑	MOS ↑	Noisiness ↑	PESQ ↑	STOI ↑
GSV	97.40%	97.80%	67.20%	2.68 ± 0.06	3.50 ± 0.04	2.22 ± 0.03	0.88 ± 0.00
F-VC	99.80%	99.20%	97.40%	3.38 ± 0.06	2.72 ± 0.06	2.22 ± 0.02	0.91 ± 0.00
Q-VC	99.40%	99.00%	97.40%	3.42 ± 0.06	2.88 ± 0.07	2.22 ± 0.02	0.93 ± 0.00
T-VC	51.60%	51.60%	20.40%	2.84 ± 0.06	3.54 ± 0.04	2.52 ± 0.03	0.86 ± 0.01

derscoring the effectiveness of our approach. For TriAAN-VC, however, adjusting the quality control weight has the opposite effect. This may be attributed to the distinct nature of the TriAAN-VC speaker encoder, which produces variable-length speaker embeddings. Such representations make loss-based quality control less effective, thereby highlighting STM’s superiority in enforcing audio distortion tolerance.

Effect of Targeted Embedding Shift. To examine the effect of targeted embedding shift on audio quality and model performance, we modify the loss function in Equation 3 by replacing $emb_{\text{spk}}^{\text{trg}}$ with $emb_{\text{spk}}^{\text{orig}}$ and reversing the sign of each term. Table V reports the STM evaluation results with the modified loss function, showing a substantial degradation in audio quality for GSV, FreeVC, and QuickVC compared with the targeted embedding shift. This demonstrates that under our perceptibility control method, guiding perturbations toward a valid subspace of speaker embeddings improves the quality of the generated audio.

Effect of Multi-Step TV Loss. We remove the multi-step TV losses from the objective function, and the corresponding evaluation results are reported in Table VI. As shown, without TV loss the protected utterances contain abruptly varying noise. These less smooth perturbations produce noticeable artifacts in the audio, thereby reducing the practicality of the protection.

TABLE VI
EVALUATION OF STM WITHOUT TV LOSS.

Model	TitaNet ↑	E-TDNN ↑	Pyannote ↑	MOS ↑	Noisiness ↑	PESQ ↑	STOI ↑
F-VC	99.40%	99.60%	95.80%	2.74 ± 0.07	3.04 ± 0.07	2.34 ± 0.02	0.92 ± 0.00
Q-VC	99.80%	99.60%	98.20%	2.83 ± 0.07	3.00 ± 0.07	2.24 ± 0.02	0.93 ± 0.00

V. CONCLUSION

In this paper, we introduced a fast and effective method for protecting speech from voice cloning attacks. By perturbing audio in the frequency domain under the guidance of a psychoacoustic model, our approach enforces strict perceptual limits while greatly accelerating adversarial perturbation generation. Experiments on state-of-the-art voice cloning models show that our method delivers competitive protection with higher audio quality and substantially reduced processing time.

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